

OPTIONS TRADING FOR BEGINNERS

A practical guide to moneyness, option pricing, and using an options calculator

Read a contract

Understand the underlying, strike, expiration, premium, and why one contract usually represents 100 shares.

Determine moneyness

Classify calls and puts as in-the-money, at-the-money, or out-of-the-money in seconds.

Use a calculator

Turn live inputs into theoretical value, Greeks, and simple what-if scenarios before you trade.

Read this first

This handbook is for education, not personal investment advice. Options are complex, involve risk, and are not suitable for every investor. Before trading, read the OCC options disclosure document and your broker's product-specific rules.

How to use this guide

If you are brand new, read the sections in order. If you already know calls and puts, jump straight to Section 2 on moneyness and Section 4 on the calculator workflow. The goal is to build one mental model: **read the contract, classify it, price it, and plan the trade before you click buy or sell.**

Section 1

What an option is and how a contract is built.

Section 2

How to determine moneyness and why moneyness is not the same as profit.

Section 3

Intrinsic value, time value, implied volatility, and the Greeks.

Section 4

Exactly what to type into an options calculator and how to read the output.

Section 5

Short practice drills that make the concepts stick.

Section 6

A learning path from absolute beginner to early intermediate trader.

Four rules worth memorizing now

- **Calls want stock above the strike.**
- **Puts want stock below the strike.**
- **Premium is quoted per share, so total cost is usually premium x 100.**
- **Moneyness is a snapshot relative to the strike; profitability also depends on the premium you paid or received.**

Useful vocabulary

Underlying	Strike	Expiration	Premium	Assignment
The stock, ETF, or index the option is based on	The fixed buy or sell price written into the contract	The date the option stops existing	The option's price, quoted per share	What happens to a seller when the contract is exercised against them

1. Options in plain English

An option is a **derivative**: its value comes from something else, called the **underlying**. For most beginners the underlying is a stock or ETF. A **call** gives the buyer the right to buy at the strike price before expiration. A **put** gives the buyer the right to sell at the strike price before expiration. The seller takes the opposite obligation if the contract is assigned.

For listed equity options, one standard contract usually controls 100 shares, so a premium of \$2.30 typically means about \$230 per contract before fees. Some adjusted contracts differ after events such as splits or mergers, so always verify the contract specifications.

Buyer and seller roles

Position	What you get	What you owe	Risk snapshot
Long Call	Right to buy at the strike	No obligation beyond premium paid	Max loss is the premium paid
Long Put	Right to sell at the strike	No obligation beyond premium paid	Max loss is the premium paid
Short Call	Premium collected up front	May have to sell if assigned	Can be very risky if uncovered
Short Put	Premium collected up front	May have to buy if assigned	Substantial downside risk if the stock falls

Anatomy of a contract

Take the quote **XYZ Apr 19 50 Call @ 2.30**. You can read it from left to right like a sentence: the underlying is XYZ, the contract expires on April 19, the strike price is \$50, it is a call, and the premium is \$2.30 per share.

Underlying	Expiration	Strike	Type	Premium
XYZ	Apr 19	\$50	Call	\$2.30
<i>Stock or ETF</i>	<i>Exact contract date</i>	<i>Exercise price</i>	<i>Call or Put</i>	<i>Price per share</i>

Two details that matter more than beginners expect

- If your calculator asks about **style**, equity options are generally American-style while many index options are European-style. That changes when exercise is allowed.
- Your exit does not have to be exercise. Long options are often closed before expiration by selling the contract instead of using the exercise right.
- Do not assume all contracts share the same expiration rhythm. Many products have weekly contracts as well as standard monthly contracts.

2. How to determine moneyness

Moneyness tells you where the option stands **right now** relative to the strike price. It does **not** tell you whether your trade is profitable yet. To determine moneyness, ignore the premium for a moment and compare the stock price with the strike.

Use this four-step method every time:

- Identify whether the contract is a **call** or a **put**.
- Look at the current **stock price** and the **strike price**.
- Apply the call or put rule below.
- Measure how many dollars the contract is in-the-money or out-of-the-money.

Contract	In-the-money	At-the-money	Out-of-the-money	Intrinsic value
Call	Stock price > strike	Stock price = strike	Stock price < strike	$\max(0, \text{stock} - \text{strike})$
Put	Stock price < strike	Stock price = strike	Stock price > strike	$\max(0, \text{strike} - \text{stock})$

Example matrix

With the stock at \$105	95 strike	105 strike	115 strike
Call	ITM by \$10	ATM	OTM by \$10
Put	OTM by \$10	ATM	ITM by \$10

Fast memory trick

Calls want the stock price **above** the strike because the right to buy becomes valuable when the market is already higher. Puts want the stock price **below** the strike because the right to sell becomes valuable when the market is already lower.

Intrinsic value and break-even

The amount a contract is already in-the-money is called **intrinsic value**. At-the-money and out-of-the-money options have no intrinsic value. But a trade can still be unprofitable even when the option is in-the-money, because your **premium** matters too.

Keep these formulas together in your notes

Call intrinsic value = $\max(0, \text{stock price} - \text{strike})$

Put intrinsic value = $\max(0, \text{strike} - \text{stock price})$

Long call break-even at expiration = $\text{strike} + \text{premium paid}$

Long put break-even at expiration = $\text{strike} - \text{premium paid}$

Worked examples

Contract	Moneyness	Intrinsic value	Premium	Time value	Notes
45 Call @ 6.50 with stock at 50	ITM by \$5	\$5.00	\$6.50	\$1.50	ITM does not guarantee profit; break-even at expiration is \$51.50
55 Call @ 1.20 with stock at 50	OTM by \$5	\$0.00	\$1.20	\$1.20	Cheaper up front, but needs a bigger move to win by expiration
55 Put @ 6.10 with stock at 50	ITM by \$5	\$5.00	\$6.10	\$1.10	Long put break-even at expiration is strike minus premium

3. What makes an option worth more or less

Every option premium can be split into two parts: **intrinsic value** and **time value** (also called extrinsic value). Intrinsic value is what the contract is worth if you stripped away time. Time value is the extra amount traders pay because the underlying still has time to move before expiration.

Intrinsic value Exists only for in-the-money options. It is the built-in value you get from the current stock price versus the strike.	Time value The portion of premium above intrinsic value. This is what shrinks as expiration approaches.	Model inputs Underlying price, strike, time, implied volatility, rates, and dividends all influence theoretical value.
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Input or driver	What it changes	Beginner note
Underlying price	Moves calls and puts in opposite directions	Usually the most visible driver day to day
Strike price	Changes moneyness and intrinsic value	This is why nearby strikes price so differently
Time to expiration	Changes time value	Time decay speeds up as expiration gets closer
Implied volatility	Changes theoretical value without changing direction	IV is about expected size of movement, not whether price goes up or down
Interest rate and dividends	Affect model value and carry	Usually smaller effects for many short-dated equity options

One idea that trips up many beginners

Implied volatility is about the market's estimate of how much the underlying might move, not whether it will move up or down. That is why an option can become more expensive even if the stock barely moves, or cheaper even when the stock stays near the same level.

The Greeks in plain English

The Greeks are estimates, not promises. They help you see what the option is most sensitive to under a pricing model.

Greek	Plain-English meaning	What beginners usually watch
Delta	Approximate premium change for a \$1 move in the underlying	Good first check for directional sensitivity
Gamma	How much Delta can change when the underlying moves	Higher near the strike and near expiration
Theta	Estimated time decay	Long options fight this every day
Vega	Sensitivity to implied volatility	Helpful around earnings, events, and volatility changes
Rho	Sensitivity to interest rates	Usually matters more in longer-dated contracts

4. How to use an options calculator

An options calculator is a pricing and risk tool. It can estimate a **theoretical value**, show the Greeks, and in many tools it can also back out **implied volatility** if you already know the market premium. Use it as a decision aid, not as a crystal ball.

What the calculator can and cannot do

It can: estimate model value, show sensitivities, compare strikes and expirations, and run simple what-if scenarios.

It cannot: guarantee a fill, predict direction with certainty, or make a risky trade safe.

Step 1: Gather the right inputs

Field	What to enter	Common beginner mistake
Underlying price	Current stock, ETF, or index price	Using an old close instead of a live quote
Call or Put	The exact contract type	Mixing up bullish calls and protective or bearish puts

Field	What to enter	Common beginner mistake
Strike	The specific exercise price	Looking at the wrong strike or the wrong month
Expiration / DTE	Exact expiration date or days to expiration	Forgetting weeklies or thinking all contracts expire the same day
Premium or IV	Use market premium to solve for IV, or use IV to estimate value	Confusing implied volatility with historical volatility
Interest rate / dividends	Model assumptions when the calculator asks for them	Leaving unrealistic defaults without checking
Style / settlement	American or European if the tool asks	Assuming all options exercise the same way

Step 2: Run the calculator the same way every time

1. Pull up the exact contract Match the underlying, type, strike, and expiration before doing any math.	2. Decide your goal Do you want theoretical value, implied volatility, or a risk comparison across strikes?	3. Enter assumptions Add current price, IV or premium, and any rate or dividend inputs the tool requests.
4. Calculate Generate theoretical value and Greeks. Write down the result before changing anything.	5. Compare with the market Check the live bid and ask, not just the last trade, because execution happens near the current market.	6. Change one variable at a time Move stock price, IV, or time one step at a time so you can see what is really driving the contract.

What to read first on the output screen

- **Theoretical value:** the model's estimate under your assumptions.
- **Delta:** your first look at directional sensitivity.
- **Theta:** how much time decay may be working against long options.
- **Vega:** how much volatility changes matter.
- **Bid/ask spread:** your reality check for trade execution.

When model value and market value disagree

Start by checking the inputs before you assume the market is wrong. A mismatch often comes from the wrong expiration, the wrong strike, stale prices, or a different volatility assumption. Also remember that different models can produce slightly different theoretical values, especially around time decay and American-style exercise features.

5. A beginner practice routine

You do not need a live trade to learn a lot. A simple paper exercise will teach you more than jumping from one social-media example to another.

Ten-minute drill

- Choose one liquid stock or ETF and write down the current price.
- Pick three strikes around the stock price: one below, one near, and one above.
- Label each call and put as ITM, ATM, or OTM.
- Estimate intrinsic value before you open the calculator.
- Run the calculator and record theoretical value, Delta, Theta, and Vega.
- Then change only one variable: stock +\$1, stock -\$1, IV +1 point, or one week less to expiration.
- Write one sentence about what changed and why.

Worksheet idea

Contract	My moneyness label	My intrinsic value	Calculator value	What changed most?
Call below stock price				
Nearest strike call				
Call above stock price				
Put below stock price				
Nearest strike put				

Common beginner mistakes

Mistake 1

Confusing in-the-money with profitable. The premium you paid still matters.

Mistake 2

Forgetting the 100x multiplier and underestimating position size.

Mistake 3

Looking at the last trade instead of the current bid/ask spread.

Mistake 4

Ignoring assignment and expiration rules on short options.

Mistake 5

Treating implied volatility as a directional forecast.

Mistake 6

Holding into expiration without knowing your plan to close, exercise, or let the option expire.

6. From absolute beginner to early intermediate

A strong foundation in options is not about memorizing strategies. It is about understanding risk, contract mechanics, and how price, time, and volatility interact.

Stage	What to learn	What you should be able to answer first
1. Beginner	Calls, puts, moneyness, intrinsic vs time value, order entry, assignment basics	Can I explain the contract in one sentence and state my max loss?
2. Early intermediate	Defined-risk spreads, Delta/Theta tradeoffs, IV changes, expiration planning	Do I know why I picked this strike and why this expiration fits the idea?
3. Product-specific detail	Index vs equity differences, settlement style, event volatility, portfolio use, advanced adjustments	Do I understand the product rules well enough to avoid accidental risk?

A conservative learning order

- Start with **long calls and long puts** so the maximum loss is easier to see.
- Learn how **protective puts** and **covered calls** relate to existing stock positions.
- Only move into multi-leg positions after you can explain moneyness, break-even, and the Greeks without looking them up.
- Do not treat uncovered short options as a beginner strategy. Those positions can carry very large losses.

7. Pre-trade checklist

Before every live trade, try to answer each line below in one sentence.

Question	Why it matters
What is my thesis: bullish, bearish, hedge, or income?	The contract should match the job you want it to do.
What is the exact contract: underlying, type, strike, expiration, and size?	This avoids buying the wrong month or the wrong strike.
What is my max loss and my break-even at expiration?	You need both numbers before placing the order.
What will I do before expiration: close, exercise, or let expire?	A plan reduces last-minute decision errors.
Am I using the bid/ask spread and a sensible order type?	Execution price can matter as much as your market idea.
Do I understand assignment risk if I am short options?	Sellers have obligations, not just premium income.

Question	Why it matters
Have I read the OCC disclosure and my broker's rules?	Approval, deadlines, and product features vary by firm and product.

Quick reference sheet

Keep this page handy when you are reading an option chain or working inside a calculator.

Concept	Remember this
Call	Right to buy at the strike
Put	Right to sell at the strike
Call ITM	Stock price above strike
Put ITM	Stock price below strike
Intrinsic value	Amount already in-the-money
Time value	Premium above intrinsic value
Long call break-even	Strike + premium paid
Long put break-even	Strike - premium paid
Calculator first look	Theoretical value, Delta, Theta, Vega, and the bid/ask spread

One final mental model

Read the contract first. Then classify moneyness. Then estimate intrinsic and time value. Then use the calculator to see how price, time, and volatility may change the trade. If you cannot explain those four steps in plain English, the trade is probably too complex for now.

Source notes and official learning resources

Use the current versions of these official resources for product rules, disclosures, and platform details.

OIC - Options Basics and Learning Center

OIC - Options Pricing and Greeks

Cboe - Options Calculator

SEC Investor.gov - An Introduction to Options

SEC Investor.gov - Types of Orders

FINRA - Options

OCC - Characteristics and Risks of Standardized Options